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REVIEW ARTICLE

Carbohydrate Structure Database and Other Glycan Databases as an Important Element of Glycoinformatics

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Abstract—Carbohydrates are one of the most chemically diverse classes of biomolecules. The amount of accumulated information on carbohydrates is far beyond the level allowing navigation in this data ocean without special tools, which are glycomic databases and prognostic services built on top of these data. Existing databases, focused on solving the particular challenges in glycoscience, are not fully compatible with each other in coverage, data formats, and features served to users. Major problems in the modern glyco-databases include data quality, gaps in coverage, and absence of a widely accepted carbohydrate notation. Most demanded are databases with broad coverage, which can provide a universal dataspace on structures, properties, and functions of carbohydrates, associated with taxonomy and other features of their natural sources. In the framework of the Carbohydrate Structure Database (CSDB) project, we created a database architecture aimed at development of the extensible glycoinformatic portal with continuous maintenance and regular content updates. This architecture was implemented in software free of drawbacks typical for glycomic databases. For the 15 years of existence, CSDB has become the main source of data on glycans of microorganisms, and a platform for multiple carbohydrate-related services. This project includes a global-scale database of natural carbohydrates; among its key features are free access, annual data deposition and updates, search and correction of errors (including those in publications), and regular announcement of new services.

Keywords: CSDB, carbohydrates, databases, glycoinformatics

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INTRODUCTION

Carbohydrates are one of the most chemically diverse classes of biomacromolecules. After the discovery of protein glycosylation and clarification of the role of carbohydrate antigens in intercellular interactions, the interest in them has been constantly growing. Nowadays, orientation in currently acquired scope of glyco-related information is impossible without special tools of informatics. Therefore, the progress of glycobiology strongly depends on the availability of an informatic environment, including data on structure, properties and functions of carbohydrates, linked to taxonomy and biological sources. The main approach to create such an environment are glycomic databases and prognostic services on top of them. Unlike genomics and proteomics, structure identification standards and data exchange protocols in glycomics were standardized only in recent years; this process is not yet fully completed. The emerging projects in the new field of bioinformatics (glycoinformatics) are not completely compatible with each other in coverage, data formats, and services provided to chemists, biologists, geneticists, and pharmacists. Each of these projects is devoted to its specific tasks; however, there is a clear trend towards mutual integra-

Abbreviations: API, application programming interface; DB, database; CCSD, complex carbohydrate structure database; CSDB, Carbohydrate Structure Database; ESKAPE, Enterococcus faecium Staphyllococcus aureus Klebsiella pneumonia Acinetobacter baumannii Pseudomonas aeruginosa Enterobacter spp.; IUPAC, International Union of Pure and Applied Chemistry; NCBI, National Center for Biotechnology Information; PDB, Protein Data Bank; SNFG, Symbol Nomenclature for Glycans.

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tion. A more detailed analysis of problems in glycoinformatics and their possible solutions has been published as a mini-review [1].

EXISTING GLYCOMICS DATABASES

Glycomics databases can be divided into structural, proteomic, functional, and auxiliary databases. The most valuable databases focused on carbohydrate structures are presented in Fig. 1.

Carbohydrate databases with a wide coverage are most demanded, among them: GLYCOSCIENCES [2–4] (CCSD/CarbBank import [5, 6] + mammalian carbohydrates + NMR data), UniCarbKB [7, 8] (mammalian O- and N-glycans, including GlycoSuite [9, 10] and Glycobase/GlycoStore (NIBRT) [11, 12] data), KEGG Glycan [13] (mainly CCSD/CarbBank import), and Carbohydrate Structure Database [14–17] (CSDB; carbohydrates of prokaryotes, plants, and fungi + NMR data). Specialized databases should be also noted: GlyGen portal [18, 19] (structural and proteomic glycan database, which has inherited (similar to UniCarbKB) the GlycoSuite data and the architecture of EuroCarbDB design project [7, 20]), ECODAB [21] (*Escherichia coli* O-antigens), MonosaccharideDB [22, 23] (monosaccharides), GlyTouCan (a meta-repository of structures [24, 25] created as a source of universal carbohydrate identifiers), and GlycoBase-Lille (carbohydrates of amphibians and other animals + NMR data). Historically, CCSD/CarbBank was the first universal carbohydrate DB [6], claiming completeness of coverage for all structures published before 1996, when its support was ceased. Since the collection and digitization of primary data from publications is the most labor-intensive part of a database development, almost all modern projects somehow use the CarbBank data, as well as elements of its ideology.

GLYCAN DATABASE ESTIMATION CRITERIA

Distinctive features, as well as criteria for the evaluation of carbohydrate databases, include the types of deposited data, completeness of coverage, data quality, functionality (as well as stability and performance), user interface, integration with other projects and, implicitly, internal architecture of a DB.

The types of information, the storage and processing of which are necessary for a carbohydrate database, include, as a minimum, the primary structure of molecules, their taxonomic, and bibliographical annotations. DBs frequently include experimental data, for example, NMR or mass spectra. The storage of biochemical, genetic, medical and other related data is often supported, but their coverage remains poor. Taxonomic and bibliographic annotations are also available in not all the databases or not for all records. In databases providing the NMR spectra, the NMR coverage is 5–35% of the structures. In Fig. 1, the main types of the deposited information are indicated by icons. For primary databases that receive data from their own annotation efforts, the icons are marked with ORIG; secondary databases import the structures from other projects, supplementing them with derived data.

The completeness of coverage significantly increases the DB value, since in this case even a negative response to a search query presents valuable scientific information. The completeness of coverage is limited by the inability to automate the search process for articles with primary data. At present, only bacterial and fungal parts of CSDB claim a full (>90%) coverage within a chosen domain. The coverage remains relevant with a timely update of the database; an acceptable time lag between publication and deposition is ~1–2 years. The requirement of journals' editorial boards to obligatorily upload all described structures to databases prior to publication (with subsequent assignment of a record ID) is a universal solution to improve the relevance of data. Such approach has been long ago implemented in genomics, but is absent in glycomics due to insufficient standardization of glycan description languages (notations) originating from high chemical diversity of carbohydrates.

The process of data deposition can hardly be automated not only at the level of publication selection, but also at the level of article interpretation. As a consequence, all chemical and biological DBs contain errors (listed in population-descending order): introduced by annotators, transferred from other DBs, present in the original publications, emerged due to DB architecture inconsistency and software bugs in importers and auto-annotators. According to the results of our dedicated study, most records in CarbBank contain errors [26]; moreover, more than a third of records contain two or more errors, the most frequent of which is incorrect taxonomic annotation of the structure. Significant gaps in the completeness of coverage were also found. Since many modern projects use CarbBank data, these errors are continuously reproduced. Some types of errors can be detected (and sometimes corrected) automatically, and such a control is carried out in several projects; however, a retrospective expert analysis of publications and curation by glycobiologists are required to achieve a high data quality.

The functionality of a DB is its capability to process search queries of different types, combine them using different logical operations, and refine them using data of other types. For example, "find all structures published in 2001–2005 containing a given fragment, as well as a monosaccharide-bound lysine or alanine, other than synthetic ones or those found in gammaproteobacteria, then display their NMR spectra." The carbohydrate-related services (generation of conformational maps, prediction of spectra, search for struc-

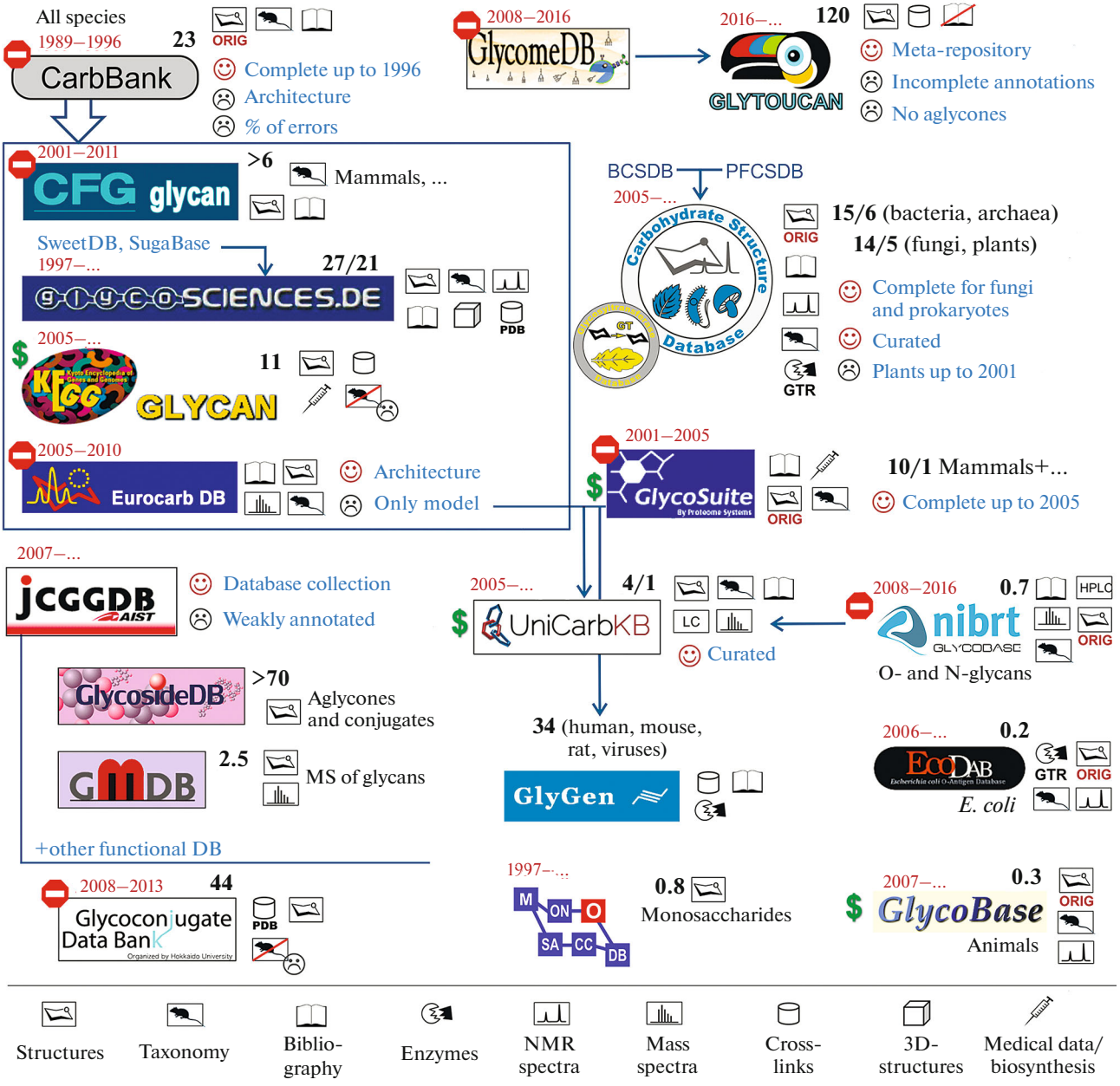


Fig. 1. Review of existing structure–centric carbohydrate databases. Icons reflect the main types of accumulated data according to the designations at the bottom of the figure. Arrows indicate the data reuse. The approximate coverage in structures/publications (in thousands of records) is indicated by black numbers. The main advantages and disadvantages are listed in blue font. The projects with paid access are marked by a green symbol $. The ceased projects, which influenced the existing ones, are designated by a red octagon with a white line. The figure was adapted from the work of Egorova and Toukach (2018) [1].

tural regularities, statistical processing and clustering of data, etc.) also refer to functionality.

Figure 2 shows the main user requests which allow transitions from one type of carbohydrate data to another. There are generally recognized standard indices for many types of data. GlyTouCan ID can serve as such an index for bioglycan primary structures. Unlike easy-to-implement search schemes for bibliography, key words, text fragments, taxonomy, etc., a search for partially matching substructures, as well as a search for the structures and spectra "similar" to those indicated, is a problem requiring preliminary studies, sophisticated programming, and significant computing resources. In this regard, of significance is the internal DB architecture, whose design is critical for achievement of reasonable processing rates of structural queries. At the stage of the development of glycoinformatics in the 2010s, the research team GLYCOSCIENCES.de formulated

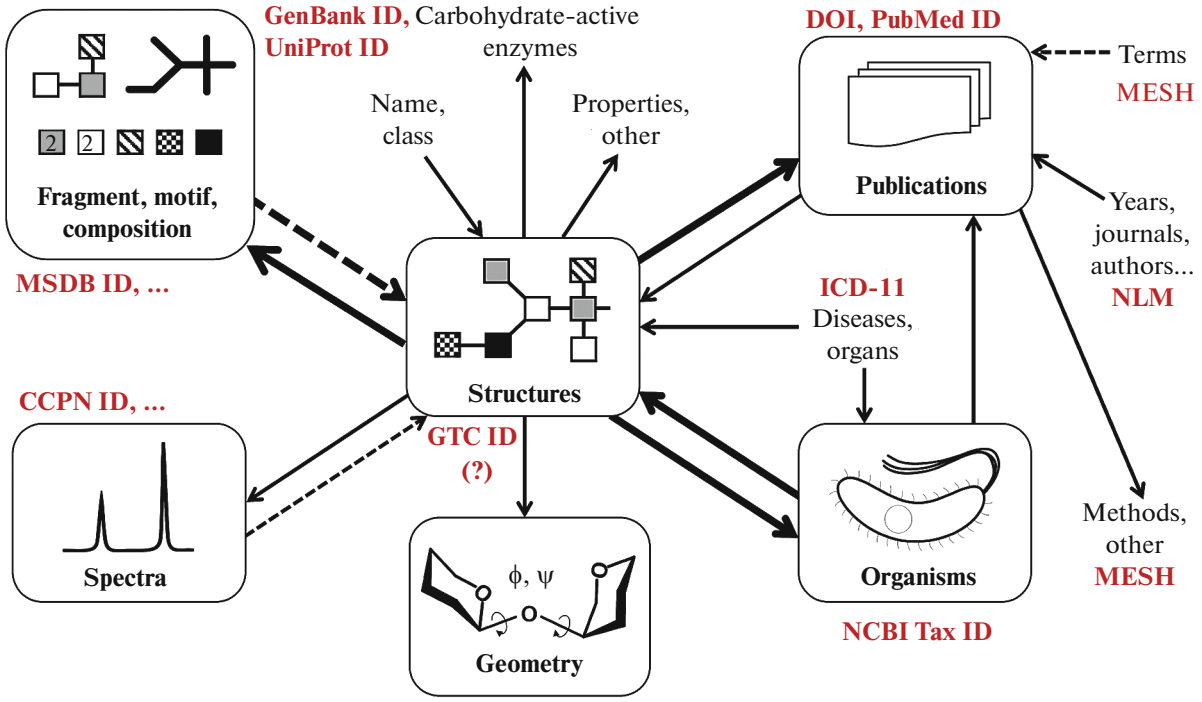


Fig. 2. Relationship between the types of information presented in the Carbohydrate Structure Database (CSDB). Unambiguous queries and data flows are depicted by solid arrows; the ones implying the fuzzy logic (identification of patterns by analogy or partial match) are depicted by dotted arrows. Thick arrows correspond to the most common user requests. Any queries can be combined using Boolean operations AND, OR, and NOT.

the "Ten golden rules of carbohydrate database development" that combined the experience of the German and Russian groups and that were implemented in the carbohydrate notation GlycoCT [27] and design project GlycomeDB [28] (GlyTouCan precursor). The key points of this document included usage of a connection table for inward structure representation, maximal possible indexing, minimum of the free text data (which, regretfully, are present in every project), and controlled vocabularies for as much data types as possible (primarily, for the names of residues). An attempt to separate the dictionary of monomers from the area of responsibility of specific projects was made within the MonosaccharideDB database [22]. Further improvement of these rules by the Glycoinformatics Consortium (https://glic.glycoinfo.org) and the Glycan Informatics Advisory Group (https://www.ncbi.nlm.nih.gov/ glycans/glyag.html) included standardization of glycan presentation in publications and on the web (SNFG [29, 30]), and a trend to use the semantic web (with Resource Description Framework) for obtaining the inferred database-independent knowledge [32–34]. The adaptation of this model to the chemistry and biology of carbohydrates was expressed in the ontologies GlycoRDF (common) [35] and GlycoCoO (glycoconjugates) [36].

CARBOHYDRATE NOTATIONS

The possibility of correct processing of structural information is closely related to the way of recording carbohydrate structures [37]. The imperfection and incompatibility of the formats of these records were for a long time a cornerstone for the progress of glycoinformatics. Languages (notations) used for internal representation of data and/or for user interface are evaluated according to the following criteria:

(1) Unambiguity (strict rules for recording each chemically distinct structure in a unique way).

(2) Ability to process the maximum possible part of naturally occurring structures (polymeric, oligomeric, and combined carbohydrates, glycolipids, glycoproteins), including those with non-carbohydrate components, and with the support for all kinds of "special cases" (nonstandard residues, bonds via phosphorus and sulfur, cyclic ethers, amide and ester bonds, etc.).

(3) The ability to work with incomplete (partially determined) structures at the level of residues, their configurations, substitution positions, structure topology, and stoichiometry of the side chains).

(4) Computer-readability (without the need for complex parsing as, for example, in the case of Extended IUPAC notation) and human-readability (required for tracking errors that inevitably occur during "human" processing of data), including the names of residues understandable to glycobiologists.

(5) Compatibility with the existing formats and atomic models (the presence of converters facilitating language learning and transition between DBs).

(6) Independence from manually curated resources, such as vocabularies of monomers, ligands, etc.

At present, CSDB Linear [37], GlycoCT [27], and WURCS [38, 39] notations provide the indicated characteristics to the greatest extent. However, the former does not support some topologies, while the latter two are not human-readable. In contrast to genomics and proteomics, glycomics still lacks a widely recognized notation except the IUPAC, highly imperfect for this application. One of the mentioned languages can become such a standard in the future.

DATABASE INTERFACE AND INTEGRATION

Modern quality standards in scientific software engineering imply that the user interface (as well as administrator interface) is intuitive, well-documented, and available free of charge to the scientific community via the Internet. Intuitiveness includes structure input and output formats, which users should not have to study. In this aspect, the implementation of input of structure fragments using independent services is extremely productive. This includes special editors with a programming interface allowing any DB to use the interface of other DBs. A brief overview of modern carbohydrate editors was published in the article devoted to CSDB/SNFG editor [40].

Integration between glycoinformatics projects implies not only a common interface for search queries, but also possibility of the automated data exchange. This concerns the interaction with non-carbohydrate databases: bibliographic (for example, NCBI PubMed) [41], taxonomic (NCBI Taxonomy) [42], genetic (NCBI GenBank) [43], proteomic (UniProt) [44], etc. GLYCOSCIENCES.de and Bacterial CSDB were the first projects that developed the protocols for automatic exchange of carbohydrate data [45], after which the standardization of formats and the development of glycomics software services accelerated significantly.

EuroCarbDB [20] stands apart; it was funded as a DB completely devoid of shortcomings and providing the widest functionality at a reasonable cost of service, but in fact was limited to the development of approaches (without their real implementation, which is a "bottleneck" of DBs due to the human factor) and CCSD/CarbBank import. The opposite end of the ideological hierarchy is occupied by the GlyTouCan repository, which does not provide own data but integrates with many other projects by importing their data and acting essentially as a meta-database, which allows cross-project operations in a single interface.

A deeper analysis of the state of glycoinformatics and its modern initiatives, carbohydrate databases, and accompanying projects can be found in the reviews and collections [46–51].

CARBOHYDRATE STRUCTURE DATABASE

Within the Carbohydrate Structure Database (CSDB) project, we aimed at the design of a DB architecture and its implementation in a software product, which would be devoid of the main shortcomings of other DBs of glycomics, as well as at provision of the continuous support, which does not require significant financial investments, and regular updates. The completeness of coverage and completely verifiable content are key features of the CSDB. During 15 years of its existence, the CSDB has become the main source of data on carbohydrates of microorganisms, and a platform for many carbohydrate-related services. The project is aimed at creating a modern and comprehensive database of natural carbohydrates, which can ideologically replace CarbBank.

The CSDB team carries out systematic work on informatization of glycomics [1, 52]. In collaboration with the global community of glycoinformaticians, criteria for the quality of programs and services in this field were formulated, and standards and ontologies for computer representation and visualization of carbohydrate data were created. The CSDB platform includes thematic databases and calculation modules that allow drawing conclusions based on the processing of all data, including screening and statistical processing. The main features and screenshots of selected database tools are presented in Fig. 3. All functions of the project are available for free to glycochemists and glycobiologists via the Internet (http://csdb.glycoscience.ru). The major milestones of its development by year are given at the project website in the "History of changes" section (http://csdb.glycoscience.ru/help/about.html#changelog). Particularly, innovations in 2020–2021 include the integration with International Classification of Diseases (ICD-11) and KEGG database, a 3D modeling module, SNFG-compatible graphical structure editor, an oligosaccharide conformation module, and service subdatabases for aglycones and glycoepitopes.

The essential components of CSDB are listed below.

(1) Database of natural carbohydrates of bacteria, archaea, fungi, plants, and protozoa [15, 16, 53]. For prokaryotes and fungi, the database provides close to complete coverage i.e., it includes ~90% of all published data, which makes even a negative response to a search query a scientifically valuable answer). The database contains data on the primary structure of glycans and glycoconjugates, their taxonomic reference down to strains and serogroups, detailed bibliographic annotations, NMR spectra and partly biochemical, genetic, medical and other annotations, as well as references to other databases (NCBI PubMed [41], NCBI Taxonomy [42], ICD-11 [54], MeSH [55], PubChem [56], GlyTouCan [24, 25], Thomson Reuters DCI [57]). Data enter the database based on publication search in bibliographic databases and subse-

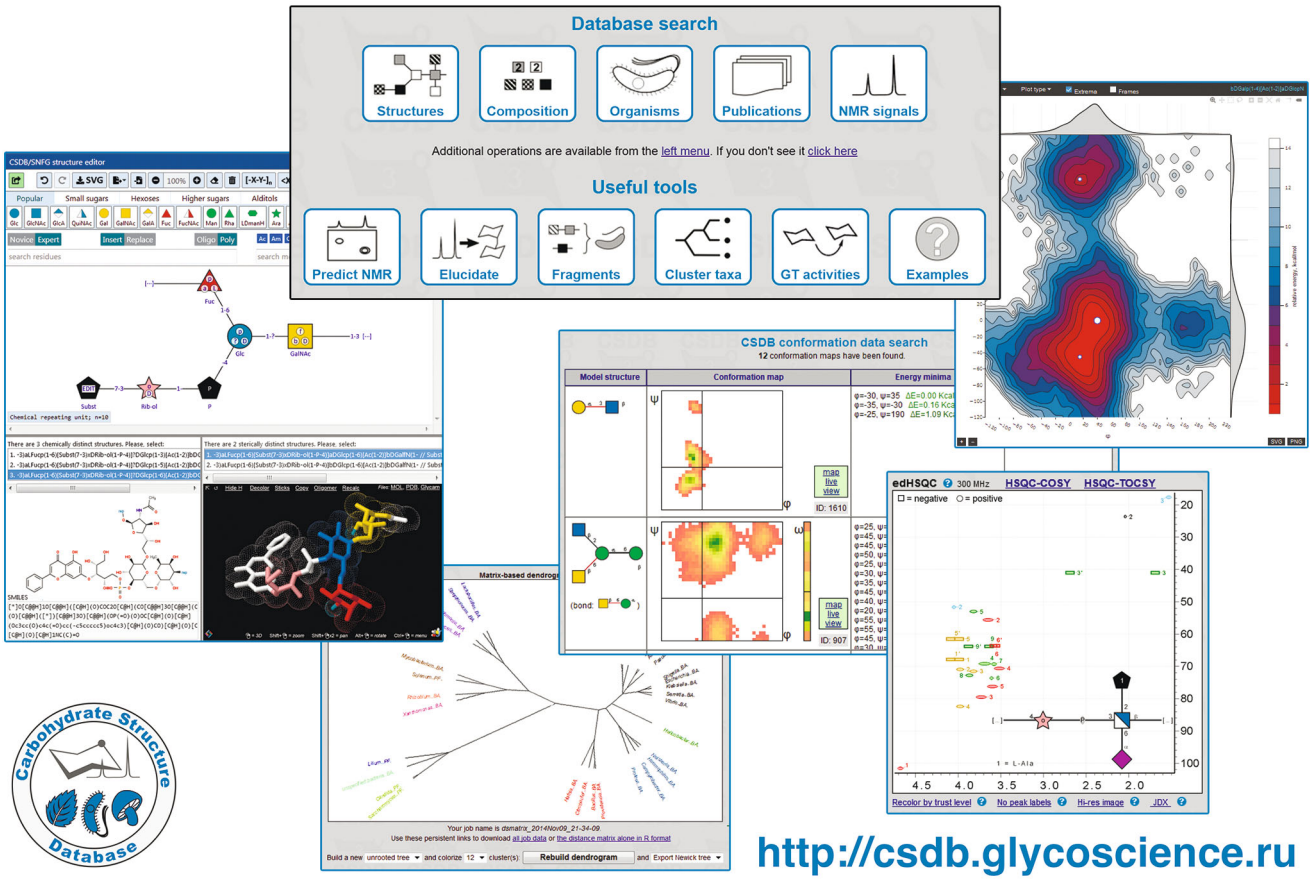


Fig. 3. The screenshots of selected Carbohydrate Structure Database (CSDB) tools. Middle top, entry page of CSDB web portal with links to the main tools: top row, search by structure fragment, composition, organism, publication, NMR data; lower row, NMR spectra prediction service, NMR-based structure prediction, structural fragment distribution analysis, clustering of taxa by their glycomes, activity of glycosyltransferases, examples of use. Lower row, left to right screenshots: CSDB/SNFG structure editor, clustering of taxa, conformational data of disaccharides, predicted 2D HSQC spectrum, interactive conformational map.

quent article annotation. The interval between publication and deposition in the database is ~1 year.

(2) Database of conformations of glycosidic bridges in oligosaccharides and related structural fragments filled with low-temperature molecular dynamics data with explicit solvent [58] and with data imported from GlycoMapsDB database [59]. The database interface permits to study, visualize, and export energy and population maps with a dimension of up to 4.

(3) Database of glycosyltransferases with experimentally confirmed activity and links to complete structures and strains. This module provides a close to complete coverage up to 2020 for the most studied members of the three kingdoms: the bacterium *E. coli*, the yeast *Sacharomyces cerevisiae,* and the plant *Arabidopsis thaliana* [60–62].

(4) Database of structural components of natural carbohydrates (monosaccharides, alditols amino acids, fatty acids, widespread aglycones, and other molecular building blocks) with detailed structural and chemical annotations and 3D models.

(5) Numerous tools for search, filtering [63, 64], comparison, statistical processing [65], and clustering of structural, taxonomic, biosynthetic, conformational, bibliographic, NMR spectroscopical, and other data on miltiple criteria.

(6) Visual online editor of carbohydrate structures [40], which allows assembling of complex biomolecules by intuitive operations in a browser. The editor is able to export results to all modern glycoinformatics formats, to atomic formats (MOL, PDB), structural formulas, three-dimensional models, and it its own generator of possible isomers for the structures with uncertainties. Its geometry generator and visualizer [66] are optimized for the molecular mechanics of complex carbohydrates and based on a prefilled service database of conformations of monomeric residues.

(7) Module for the prediction of one- and two-dimensional NMR spectra of arbitrary carbohydrates

[67] and for structure predictions from the NMR spectra [68] and data of other experiments. The prediction of spectra is based on our own theory [69] and using the database, and provides an average accuracy for the structural components of bioglycans at the level of 0.07 ppm (^{1}H) and 0.6 ppm (^{13}C). For complex carbohydrates and glycoconjugates, the achieved accuracy significantly exceeds that of analogs, including neural network and quantum mechanics calculations at high levels of theory [70], while being 1000–10000 times faster. Such performance renders this tool suitable for bulk generation of spectra and their automatic comparison with the experimental spectra.

(8) Integration with other projects [34] of glyco-, chemo-, and bioinformatics at the level of software interface, cross-links, universal identifiers, RDF ontology [35], agreements on standards and data formats. The native carbohydrate notation used (CSDB Linear language [37]) supports the vast majority of structural features of bioglycans, including unambiguously describing their non-carbohydrate components. It has translators into other carbohydrate notations (GlycoCT, WURCS, Sweet-DB, SNFG) and back (GlycoCT).

(9) Tools for automating the annotation of original publications, detection of errors in publications and databases, programs for increasing the efficiency and reducing the cost of retrospective analysis of publications by the CSDB team.

(10) Detailed help for users, administrators, and annotators, including user manual ("Help" section in the main menu of the project site), tutorial on solving the common problems [64] (http://csdb.glycoscience.ru/help/examples.html), reference data for developers (http://csdb.glycoscience.ru/help/dbdocs.html), and quick context help for interface elements.

The CSDB is updated and supplemented with data annually, search and elimination of errors are performed (including in publications of other authors), and new services are emerging. The planned further development of the CSDB includes the expansion of coverage with animal glycans, achievement of complete coverage for protozoa, formalization of the search for glycoepitopes, improvement of the conformational module and further completion of its service database with data on branched trisaccharides, extension of glycosyltransferase database coverage with the enzymes of ESKAPE group pathogens, and other tasks.

CSDB TECHNICAL IMPLEMENTATION

CSDB is technically a platform for computer services, hosting a database and an original API. Data imported from other databases (CarbBank, GlycomeDB, GlycoMapsDB, GlycoEpitopeDB) and obtained from literature are converted after a comprehensive review and validation to human-readable text files (dumps), simultaneously acting as backups. All database content edits occur at the level of these files. Dumps are imported annually into a relational database on MySQL 5 database engine, where they obtain additional representations which accelerate the multicriterial search: connectivity table at the level of monomeric residues, service databases, and denormalization. The scripts for importing, processing, searching for data, and maintaining the database are written in PHP 5.6; the work with atomic chemical models in Python 3.5 + RDKIT + openBabel; web user interface in Javascript + DHTML. Tinker 8.3 is used for molecular dynamics calculations. The project is deployed on a dedicated hardware web server (64 cores, 96 GB RAM) managed by Windows Server 2016; it is also used as a computational server for processing of resource-intensive user queries and background calculation of conformations. Long running tasks (for example, detection of errors in dumps prepared by annotators or search for structural hypotheses best matching the experimental data) are executed asynchronously with user notification as soon as the results are ready.

CONCLUSIONS

The efficient work with the currently accumulated amount of information on carbohydrates requires an open information space of data on structure, properties, and functions of carbohydrates, associated with the taxonomy and properties of their natural sources. Glycomics databases (DB) and predictive services using data from these databases are the main means of creating such a space. For the last decade, informatization of glycochemistry and glycobiology has grown significantly, generally recognized standards and services has emerged, their compatibility with each other has increased, and the establishment of interaction with computer projects in related areas has started. This gradually brings glycomics closer to the level of informatization existing in genomics and proteomics, despite a great variability and heterogeneity of the objects of study.

The Carbohydrate Structure Database (CSDB) team conducted an extensive study on existing glycoinformatics initiatives to identify their advantages and disadvantages. The information obtained formed the basis for the creation of the DB architecture and its implementation in the CSDB software product, for which the completeness of coverage and completely verifiable content are key features. The CSDB contains data on the primary structure of glycans and glycoconjugates, their taxonomic linkage down to strains, detailed bibliographic annotations, NMR spectra and partly biochemical, genetic, medical, and other annotations. The built-in addons allow analyzing the biosynthetic pathways through glycosyltransferase database, obtaining conformational maps of oligosaccharides, simulation and assignment of the ^{13}C, ^{1}H, 2D

NMR spectra, prediction of the structure from the NMR and other data, taxon clustering based on glycomes, analysis of the fragment distribution by taxa and position in structures, classification of monomers, etc. Data in the database are annually updated, search and elimination of errors are carried out, and new services appear.

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COMPLIANCE WITH ETHICAL STANDARDS

This article does not contain descriptions of any studies involving human participants or animals as objects of studies.

Conflict of Interests

The authors declare that they have no conflict of interest.

REFERENCES

1. Egorova, K.S. and Toukach, P.V., *Angew. Chem., Int. Ed. Engl.*, 2018, vol. 57, pp. 14986–14990. https://doi.org/10.1002/anie.201803576
2. Lütteke, T., in *A Practical Guide to Using Glycomics Databases*, Aoki-Kinoshita, K.F., Ed., Tokyo: Springer, 2017, pp. 335–350. https://doi.org/10.1007/978-4-431-56454-6_16
3. Bohm, M., Bohne-Lang, A., Frank, M., Loss, A., Rojas-Macias, M.A., and Lütteke, T., *Nucleic Acids Res.*, 2019, vol. 47, pp. D1195–D1201. https://doi.org/10.1093/nar/gky994
4. Lütteke, T., Bohne-Lang, A., Loss, A., Goetz, T., Frank, M., and Lieth, C.W., *Glycobiology*, 2006, vol. 16, pp. 71R–81R. https://doi.org/10.1093/glycob/cwj049
5. Doubet, S., Bock, K., Smith, D., Darvill, A., and Albersheim, P., *Trends Biochem. Sci.*, 1989, vol. 14, pp. 475–477. https://doi.org/10.1016/0968-0004(89)90175-8
6. Doubet, S. and Albersheim, P., *Glycobiology*, 1992, vol. 2, pp. 505–507. https://doi.org/10.1093/glycob/2.6.505
7. Campbell, M.P., Peterson, R., Mariethoz, J., Gasteiger, E., Akune, Y., Aoki-Kinoshita, K.F., Lisacek, F., and Packer, N.H., *Nucleic Acids Res.*, 2014, vol. 42, pp. D215–D221. https://doi.org/10.1093/nar/gkt1128
8. Campbell, M.P. and Packer, N.H., *Biochim. Biophys. Acta*, 2016, vol. 1860, pp. 1669–1675. https://doi.org/10.1016/j.bbagen.2016.02.016
9. Cooper, C.A., Joshi, H.J., Harrison, M.J., Wilkins, M.R., and Packer, N.H., *Nucleic Acids Res.*, 2003, vol. 31, pp. 511–513. https://doi.org/10.1093/nar/gkg099
10. Cooper, C.A., Harrison, M.J., Wilkins, M.R., and Packer, N.H., *Nucleic Acids Res.*, 2001, vol. 29, pp. 332–335. https://doi.org/10.1093/Nar/29.1.332
11. Zhao, S., Walsh, I., Abrahams, J.L., Royle, L., Nguyen-Khuong, T., Spencer, D., Fernandes, D.L., Packer, N.H., Rudd, P.M., and Campbell, M.P., *Bioinformatics*, 2018, vol. 34, pp. 3231–3232. https://doi.org/10.1093/bioinformatics/bty319
12. Campbell, M.P., Royle, L., Radcliffe, C.M., Dwek, R.A., and Rudd, P.M., *Bioinformatics*, 2008, vol. 24, pp. 1214–1216. https://doi.org/10.1093/bioinformatics/btn090
13. Aoki-Kinoshita, K.F. and Kanehisa, M., in *Glycoinformatics*, Lütteke, T. and Frank, M., Eds., New York: Humana Press, 2015, pp. 97–107. https://doi.org/10.1007/978-1-4939-2343-4_7
14. Toukach, P.V. and Egorova, K.S., in *Glycoscience: Biology and Medicine*, Taniguchi, N., Endo, T., Hart, G., Seeberger, P., and Wong, C.H., Eds., Tokyo: Springer, 2015, pp. 241–250. https://doi.org/10.1007/978-4-431-54841-6_24
15. Toukach, P.V. and Egorova, K.S., *Nucleic Acids Res.*, 2016, vol. 44, pp. D1229–D1236. https://doi.org/10.1093/nar/gkv840
16. Toukach, P.V. and Knirel, Y.A., *Glycoconjugate J.*, 2005, vol. 2, pp. 216–217.
17. Toukach, P.V., *J. Chem. Inf. Model.*, 2011, vol. 51, pp. 159–170. https://doi.org/10.1021/ci100150d
18. York, W.S., Mazumder, R., Ranzinger, R., Edwards, N., Kahsay, R., Aoki-Kinoshita, K.F., Campbell, M.P., Cummings, R.D., Feizi, T., Martin, M., Natale, D.A., Packer, N.H., Woods, R.J., Agarwal, G., Arpinar, S., Bhat, S., Blake, J., Castro, L.J.G., Fochtman, B., Gildersleeve, J., Goldman, R., Holmes, X., Jain, V., Kulkarni, S., Mahadik, R., Mehta, A., Mousavi, R., Nakarakommula, S., Navelkar, R., Pattabiraman, N., Pierce, M.J., Ross, K., Vasudev, P., Vora, J., Williamson, T., and Zhang, W., *Glycobiology*, 2020, vol. 30, pp. 72–73. https://doi.org/10.1093/glycob/cwz080
19. Kahsay, R., Vora, J., Navelkar, R., Mousavi, R., Fochtman, B.C., Holmes, X., Pattabiraman, N., Ranzinger, R., Mahadik, R., Williamson, T., Kulkarni, S., Agarwal, G., Martin, M., Vasudev, P., Garcia, L., Edwards, N., Zhang, W., Natale, D.A., Ross, K., Aoki-

Kinoshita, K.F., Campbell, M.P., York, W.S., and Mazumder, R., *Bioinformatics*, 2020, vol. 36, pp. 3941–3943.
https://doi.org/10.1093/bioinformatics/btaa238

20. von der Lieth, C.W., Freire, A.A., Blank, D., Campbell, M.P., Ceroni, A., Damerell, D.R., Dell, A., Dwek, R.A., Ernst, B., Fogh, R., Frank, M., Geyer, H., Geyer, R., Harrison, M.J., Henrick, K., Herget, S., Hull, W.E., Ionides, J., Joshi, H.J., Kamerling, J.P., Leeflang, B.R., Lutteke, T., Lundborg, M., Maass, K., Merry, A., Ranzinger, R., Rosen, J., Royle, L., Rudd, P.M., Schloissnig, S., Stenutz, R., Vranken, W.F., Widmalm, G., and Haslam, S.M., *Glycobiology*, 2011, vol. 21, pp. 493–502.
https://doi.org/10.1093/glycob/cwq188
21. Rojas-Macias, M.A., Ståhle, J., Lütteke, T., and Widmalm, G., *Glycobiology*, 2015, vol. 25, pp. 341–347.
https://doi.org/10.1093/glycob/cwu116
22. Lütteke, T. and von der Lieth, C.W., *Glycobiology*, 2005, vol. 15, pp. 1209–1210.
https://doi.org/10.1093/glycob/cwj039
23. Lütteke, T., in *A Practical Guide to Using Glycomics Databases*, Aoki-Kinoshita, K.F., Ed., Tokyo: Springer, 2017, pp. 29–40.
https://doi.org/10.1007/978-4-431-56454-6_3
24. Fujita, A., Aoki, N.P., Shinmachi, D., Matsubara, M., Tsuchiya, S., Shiota, M., Ono, T., Yamada, I., and Aoki-Kinoshita, K.F., *Nucleic Acids Res.*, 2021, vol. 49, pp. D1529–D1533.
https://doi.org/10.1093/nar/gkaa947
25. Tiemeyer, M., Aoki, K., Paulson, J., Cummings, R.D., York, W.S., Karlsson, N.G., Lisacek, F., Packer, N.H., Campbell, M.P., Aoki, N.P., Fujita, A., Matsubara, M., Shinmachi, D., Tsuchiya, S., Yamada, I., Pierce, M., Ranzinger, R., Narimatsu, H., and Aoki-Kinoshita, K.F., *Glycobiology*, 2017, vol. 27, pp. 915–919.
https://doi.org/10.1093/glycob/cwx066
26. Egorova, K.S. and Toukach, P.V., *J. Chem. Inf. Model.*, 2012, vol. 52, pp. 2812–2814.
https://doi.org/10.1021/ci3002815
27. Herget, S., Ranzinger, R., Maass, K., and Lieth, C.W., *Carbohydr. Res.*, 2008, vol. 343, pp. 2162–2171.
https://doi.org/10.1016/j.carres.2008.03.011
28. Ranzinger, R., Herget, S., Lieth, C.W., and Frank, M., *Nucleic Acids Res.*, 2011, vol. 39, pp. D373–D376.
https://doi.org/10.1093/nar/gkq1014
29. Varki, A., Cummings, R.D., Aebi, M., Packer, N.H., Seeberger, P.H., Esko, J.D., Stanley, P., Hart, G., Darvill, A., Kinoshita, T., Prestegard, J.J., Schnaar, R.L., Freeze, H.H., Marth, J.D., Bertozzi, C.R., Etzler, M.E., Frank, M., Vliegenthart, J.F., Lutteke, T., Perez, S., Bolton, E., Rudd, P., Paulson, J., Kanehisa, M., Toukach, P., Aoki-Kinoshita, K.F., Dell, A., Narimatsu, H., York, W., Taniguchi, N., and Kornfeld, S., *Glycobiology*, 2015, vol. 25, pp. 1323–1324.
https://doi.org/10.1093/glycob/cwv091
30. Neelamegham, S., Aoki-Kinoshita, K., Bolton, E., Frank, M., Lisacek, F., Lutteke, T., O'Boyle, N., Packer, N.H., Stanley, P., Toukach, P., Varki, A., Woods, R.J., and Group, S.D., *Glycobiology*, 2019, vol. 29, pp. 620–624.
https://doi.org/10.1093/glycob/cwz045
31. Willighagen, E.L. and Brandle, M.P., *J. Cheminform.*, 2011, vol. 3, p. 15.
https://doi.org/10.1186/1758-2946-3-15
32. Aoki-Kinoshita, K.F., Aoki, N.P., Fujita, A., Fujita, N., Kawasaki, T., Matsubara, M., Okuda, S., Shikanai, T., Shinmachi, D., Solovieva, E., Suzuki, Y., Tsuchiya, S., Yamada, I., and Narimatsu, H., *Perspect. Sci.*, 2017, vol. 11, pp. 18–23.
https://doi.org/10.1016/j.pisc.2016.05.012
33. Katayama, T., Wilkinson, M.D., Aoki-Kinoshita, K.F., Kawashima, S., Yamamoto, Y., Yamaguchi, A., Okamoto, S., Kawano, S., Kim, J.D., Wang, Y., Wu, H., Kano, Y., Ono, H., Bono, H., Kocbek, S., Aerts, J., Akune, Y., Antezana, E., Arakawa, K., Aranda, B., Baran, J., Bolleman, J., Bonnal, R.J., Buttigieg, P.L., Campbell, M.P., Chen, Y.A., Chiba, H., Cock, P.J., Cohen, K.B., Constantin, A., Duck, G., Dumontier, M., Fujisawa, T., Fujiwara, T., Goto, N., Hoehndorf, R., Igarashi, Y., Itaya, H., Ito, M., Iwasaki, W., Kalas, M., Katoda, T., Kim, T., Kokubu, A., Komiyama, Y., Kotera, M., Laibe, C., Lapp, H., Lutteke, T., Marshall, M.S., Mori, T., Mori, H., Morita, M., Murakami, K., Nakao, M., Narimatsu, H., Nishide, H., Nishimura, Y., Nystrom-Persson, J., Ogishima, S., Okamura, Y., Okuda, S., Oshita, K., Packer, N.H., Prins, P., Ranzinger, R., Rocca-Serra, P., Sansone, S., Sawaki, H., Shin, S.H., Splendiani, A., Strozzi, F., Tadaka, S., Toukach, P., Uchiyama, I., Umezaki, M., Vos, R., Whetzel, P.L., Yamada, I., Yamasaki, C., Yamashita, R., York, W.S., Zmasek, C.M., Kawamoto, S., and Takagi, T., *J. Biomed. Semantics*, 2014, vol. 5, p. 5.
https://doi.org/10.1186/2041-1480-5-5
34. Aoki-Kinoshita, K.F., Bolleman, J., Campbell, M.P., Kawano, S., Kim, J.D., Lutteke, T., Matsubara, M., Okuda, S., Ranzinger, R., Sawaki, H., Shikanai, T., Shinmachi, D., Suzuki, Y., Toukach, P., Yamada, I., Packer, N.H., and Narimatsu, H., *J. Biomed. Semantics*, 2013, vol. 4, p. 39.
https://doi.org/10.1186/2041-1480-4-39
35. Ranzinger, R., Aoki-Kinoshita, K.F., Campbell, M.P., Kawano, S., Lutteke, T., Okuda, S., Shinmachi, D., Shikanai, T., Sawaki, H., Toukach, P., Matsubara, M., Yamada, I., and Narimatsu, H., *Bioinformatics*, 2015, vol. 31, pp. 919–925.
https://doi.org/10.1093/bioinformatics/btu732
36. Yamada, I., Campbell, M.P., Edwards, N., Castro, L.J., Lisacek, F., Mariethoz, J., Ono, T., Ranzinger, R., Shinmachi, D., and Aoki-Kinoshita, K.F., *Glycobiology*, 2021, vol. 31, pp. 741–750.
https://doi.org/10.1093/glycob/cwab013
37. Toukach, P.V. and Egorova, K.S., *J. Chem. Inf. Model.*, 2020, vol. 60, pp. 1276–1289.
https://doi.org/10.1021/acs.jcim.9b00744
38. Tanaka, K., Aoki-Kinoshita, K.F., Kotera, M., Sawaki, H., Tsuchiya, S., Fujita, N., Shikanai, T., Kato, M., Kawano, S., Yamada, I., and Narimatsu, H., *J. Chem. Inf. Model.*, 2014, vol. 54, pp. 1558–1566.
https://doi.org/10.1021/ci400571e
39. Matsubara, M., Aoki-Kinoshita, K.F., Aoki, N.P., Yamada, I., and Narimatsu, H., *J. Chem. Inf. Model.*, 2017, vol. 57, pp. 632–637.
https://doi.org/10.1021/acs.jcim.6b00650

40. Bochkov, A.Y. and Toukach, P.V., *J. Chem. Inf. Model.*, 2021, vol. 61, pp. 4940–4948. https://doi.org/10.1021/acs.jcim.1c00917
41. Lu, Z., *Database*, 2011, vol. 2011, art. ID baq036. https://doi.org/10.1093/database/baq036
42. Federhen, S., *Nucleic Acids Res.*, 2012, vol. 40, pp. D136–D143. https://doi.org/10.1093/nar/gkr1178
43. Benson, D.A., Cavanaugh, M., Clark, K., Karsch-Mizrachi, I., Lipman, D.J., Ostell, J., and Sayers, E.W., *Nucleic Acids Res.*, 2013, vol. 41, pp. D36–D42. https://doi.org/10.1093/nar/gks1195
44. The Uniprot Consortium, *Nucleic Acids Res.*, 2017, vol. 45, pp. D158–D169. https://doi.org/10.1093/nar/gkw1099
45. Toukach, P., Joshi, H.J., Ranzinger, R., Knirel, Y., and Lieth, C.W., *Nucleic Acids Res.*, 2007, vol. 35, pp. D280–D286. https://doi.org/10.1093/nar/gkl883
46. Li, X., Xu, Z., Hong, X., Zhang, Y., and Zou, X., *Int. J. Mol. Sci.*, 2020, vol. 21, p. 6727. https://doi.org/10.3390/ijms21186727
47. Abrahams, J.L., Taherzadeh, G., Jarvas, G., Guttman, A., Zhou, Y., and Campbell, M.P., *Curr. Opin. Struct. Biol.*, 2020, vol. 62, pp. 56–69. https://doi.org/10.1016/j.sbi.2019.11.009
48. Scherbinina, S.I. and Toukach, P.V., *Int. J. Mol. Sci.*, 2020, vol. 21, p. 7702. https://doi.org/10.3390/ijms21207702
49. Copoiu, L. and Malhotra, S., *Curr. Opin. Struct. Biol.*, 2020, vol. 62, pp. 132–139. https://doi.org/10.1016/j.sbi.2019.12.020
50. *A Practical Guide to Using Glycomics Databases*, Aoki-Kinoshita, K.F., Ed., Tokyo: Springer, 2017. https://doi.org/10.1007/978-4-431-56454-6
51. Aoki-Kinoshita, K.F., *Mol. Cell. Proteomics*, 2013, vol. 12, pp. 1036–1045. https://doi.org/10.1074/mcp.R112.026252
52. Toukach, F.V., Information technologies in structural glycochemistry and glycobiology, *Doctoral (Chem.) Dissertation (habilitation)*, Moscow: Zelinskii Inst. Org. Chem. Russ. Acad. Sci., 2019.
53. Egorova, K.S. and Toukach, P.V., *Carbohydr. Res.*, 2014, vol. 389, pp. 112–114. https://doi.org/10.1016/j.carres.2013.10.009
54. ICD-11: in Praise of Good Data, *Lancet Infect. Dis.*, 2018, vol. 18, p. 813. https://doi.org/10.1016/s1473-3099(18)30436-5
55. Baumann, N., *Int. J. Clin. Pract.*, 2016, vol. 70, pp. 171–174. https://doi.org/10.1111/ijcp.12767
56. Kim, S., Thiessen, P.A., Bolton, E.E., Chen, J., Fu, G., Gindulyte, A., Han, L., He, J., He, S., Shoemaker, B.A., Wang, J., Yu, B., Zhang, J., and Bryant, S.H., *Nucleic Acids Res.*, 2016, vol. 44, pp. D1202–D1213. https://doi.org/10.1093/nar/gkv951
57. Pavlech, L.L., *J. Med. Libr. Assoc.*, 2016, vol. 104, pp. 88–90. https://doi.org/10.3163/1536-5050.104.1.020
58. Stroylov, V., Panova, M., and Toukach, P., *Int. J. Mol. Sci.*, 2020, vol. 21, p. 7626. https://doi.org/10.3390/ijms21207626
59. Frank, M., Lutteke, T., and Lieth, C.W., *Nucleic Acids Res.*, 2007, vol. 35, pp. 287–290. https://doi.org/10.1093/nar/gkl907
60. Egorova, K.S. and Toukach, P.V., *Glycobiology*, 2017, vol. 27, pp. 285–290. https://doi.org/10.1093/glycob/cww137
61. Egorova, K.S., Knirel, Y.A., and Toukach, P.V., *Glycobiology*, 2019, vol. 29, pp. 285–287. https://doi.org/10.1093/glycob/cwz006
62. Egorova, K.S., Smirnova, N.S., and Toukach, P.V., *Glycobiology*, 2021, vol. 31, pp. 524–529. https://doi.org/10.1093/glycob/cwaa107
63. Egorova, K.S. and Toukach, P.V., in *A Practical Guide to Using Glycomics Databases*, Aoki-Kinoshita, K.F., Ed., Tokyo: Springer, 2017, pp. 75–113. https://doi.org/10.1007/978-4-431-56454-6_5
64. Toukach, P.V. and Egorova, K.S., in *Glycoinformatics*, Lütteke, T. and Frank, M., Eds., New York: Humana Press, 2015, pp. 55–85. https://doi.org/10.1007/978-1-4939-2343-4_5
65. Egorova, K.S., Kondakova, A.N., and Toukach, P.V., *Database*, 2015, vol. 2015, art. ID bav073. https://doi.org/10.1093/database/bav073
66. Chernyshov, I.Y. and Toukach, P.V., *Bioinformatics*, 2018, vol. 34, pp. 2679–2681. https://doi.org/10.1093/bioinformatics/bty168
67. Kapaev, R.R. and Toukach, P.V., *J. Chem. Inf. Model.*, 2016, vol. 56, pp. 1100–1104. https://doi.org/10.1021/acs.jcim.6b00083
68. Kapaev, R.R. and Toukach, P.V., *Bioinformatics*, 2018, vol. 34, pp. 957–963. https://doi.org/10.1093/bioinformatics/btx696
69. Kapaev, R.R., Egorova, K.S., and Toukach, P.V., *J. Chem. Inf. Model.*, 2014, vol. 54, pp. 2594–2611. https://doi.org/10.1021/ci500267u
70. Kapaev, R.R. and Toukach, P.V., *Anal. Chem.*, 2015, vol. 87, pp. 7006–7010. https://doi.org/10.1021/acs.analchem.5b01413

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